Von Neumann Stability Handout
Finite Difference Methods for $u_t + au_x = 0$

Rishabh Kumar

April 16, 2026

1 Fourier-Mode Test

For the linear advection equation

$$u_t + au_x = 0, \qquad a > 0, \tag{1}$$

write

$$\nu = \frac{ak}{h}. \tag{2}$$

Von Neumann stability is tested by inserting

$$v_m^n = G^n e^{\mathrm{i}m\theta} \tag{3}$$

into the finite difference scheme. The method is stable if

$$|G(\theta)| \leq 1 \qquad \text{for all } \theta. \tag{4}$$

2 Norms Used in the Numerical Evidence

At the final time $T = t_N$, define the grid error

$$e_m^N = v_m^N - u(x_m, T). \tag{5}$$

The convergence and sweep plots use the discrete norms

$$E_1 = h \sum_m |e_m^N|, \tag{6}$$

$$E_2 = \left(h \sum_m |e_m^N|^2 \right)^{1/2}, \tag{7}$$

$$E_\infty = \max_m |e_m^N|. \tag{8}$$

The first two approximate the continuous L^1 and L^2 norms:

$$E_1 \approx \|e(\cdot, T)\|_{L^1}, \qquad E_2 \approx \|e(\cdot, T)\|_{L^2}. \tag{9}$$

The maximum norm E_∞ records the largest pointwise error anywhere on the grid. It is therefore the most sensitive norm to a localized overshoot, undershoot, or oscillation.

The experimental order of convergence is computed by comparing errors on two successive meshes:

$$\mathrm{EOC} = \frac{\log(E_h/E_{h/2})}{\log 2}. \tag{10}$$

If $E_h \approx Ch^p$, then EOC $\approx p$.

For the λ-stability sweep, one also monitors

$$\left\|v^N\right\|_\infty = \max_m |v_m^N|. \tag{11}$$

This is not an error norm; it is a final-time amplitude. Since the exact solution is bounded by 1, very large values of $\left\|v^N\right\|_\infty$ indicate nonphysical growth and hence numerical instability.

3 Explicit Schemes

3.1 FTFS

The scheme is

$$v_m^{n+1} = v_m^n - \nu(v_{m+1}^n - v_m^n). \tag{12}$$

Substituting $v_m^n = G^n e^{\mathrm{i}m\theta}$ gives

$$G = 1 - \nu(e^{\mathrm{i}\theta} - 1) = 1 + \nu - \nu e^{\mathrm{i}\theta}. \tag{13}$$

Thus

$$|G|^2 = (1 + \nu - \nu\cos\theta)^2 + \nu^2 \sin^2\theta \tag{14}$$

$$= 1 + 2\nu(1+\nu)(1-\cos\theta). \tag{15}$$

For $\nu > 0$ and $\theta \neq 0$, $|G| > 1$. FTFS is unstable for right-moving advection.

3.2 FTBS

The scheme is

$$v_m^{n+1} = v_m^n - \nu(v_m^n - v_{m-1}^n). \tag{16}$$

Then

$$G = 1 - \nu(1 - e^{-\mathrm{i}\theta}) = 1 - \nu + \nu e^{-\mathrm{i}\theta}. \tag{17}$$

Therefore

$$|G|^2 = (1 - \nu + \nu\cos\theta)^2 + \nu^2 \sin^2\theta \tag{18}$$

$$= 1 - 2\nu(1-\nu)(1-\cos\theta). \tag{19}$$

For all θ,

$$|G| \leq 1 \quad \Longleftrightarrow \quad 0 \leq \nu \leq 1. \tag{20}$$

This is the usual upwind CFL condition.

3.3 FTCS

The scheme is

$$v_m^{n+1} = v_m^n - \frac{\nu}{2}(v_{m+1}^n - v_{m-1}^n). \tag{21}$$

Hence

$$G = 1 - \frac{\nu}{2}(e^{\mathrm{i}\theta} - e^{-\mathrm{i}\theta}) = 1 - \mathrm{i}\nu \sin\theta. \tag{22}$$

Consequently

$$|G|^2 = 1 + \nu^2 \sin^2\theta. \tag{23}$$

For every nonzero ν, some Fourier modes satisfy $|G| > 1$. FTCS is unstable.

3.4 CTCS or Leapfrog

The scheme is

$$v_m^{n+1} = v_m^{n-1} - \nu(v_{m+1}^n - v_{m-1}^n). \tag{24}$$

Substituting a Fourier mode gives

$$G - G^{-1} = -\nu(e^{\mathrm{i}\theta} - e^{-\mathrm{i}\theta}) = -2\mathrm{i}\nu \sin\theta. \tag{25}$$

Multiplying by G,

$$G^2 + 2\mathrm{i}\nu \sin\theta\, G - 1 = 0. \tag{26}$$

The roots are

$$G = -\mathrm{i}\nu \sin\theta \pm \sqrt{1 - \nu^2 \sin^2\theta}. \tag{27}$$

For the square root to be real for all θ, we need

$$|\nu| \le 1. \tag{28}$$

When this holds, the two roots lie on the unit circle. Leapfrog is therefore stable under $|\nu| \le 1$, but it has a computational mode and needs a stable startup step.

3.5 Lax–Friedrichs

The scheme is

$$v_m^{n+1} = \frac{1}{2}(v_{m+1}^n + v_{m-1}^n) - \frac{\nu}{2}(v_{m+1}^n - v_{m-1}^n). \tag{29}$$

Thus

$$G = \frac{1}{2}(e^{\mathrm{i}\theta} + e^{-\mathrm{i}\theta}) - \frac{\nu}{2}(e^{\mathrm{i}\theta} - e^{-\mathrm{i}\theta}) \tag{30}$$

$$= \cos\theta - \mathrm{i}\nu \sin\theta. \tag{31}$$

Therefore

$$|G|^2 = \cos^2\theta + \nu^2 \sin^2\theta = 1 - (1 - \nu^2)\sin^2\theta. \tag{32}$$

The stability condition is

$$|\nu| \le 1. \tag{33}$$

3.6 Lax–Wendroff

The scheme is

$$v_m^{n+1} = v_m^n - \frac{\nu}{2}(v_{m+1}^n - v_{m-1}^n) + \frac{\nu^2}{2}(v_{m+1}^n - 2v_m^n + v_{m-1}^n). \tag{34}$$

The amplification factor is

$$G = 1 - \mathrm{i}\nu \sin\theta + \nu^2(\cos\theta - 1). \tag{35}$$

Let $s = \sin(\theta/2)$. Then

$$\cos\theta - 1 = -2s^2, \qquad \sin^2\theta = 4s^2(1 - s^2). \tag{36}$$

Therefore

$$|G|^2 = (1 - 2\nu^2 s^2)^2 + 4\nu^2 s^2(1 - s^2) \tag{37}$$

$$= 1 - 4\nu^2(1 - \nu^2)s^4. \tag{38}$$

Hence

$$|G| \le 1 \quad \text{for all } \theta \quad \Longleftrightarrow \quad |\nu| \le 1. \tag{39}$$

3.7 Beam–Warming

For $a > 0$, the Beam–Warming scheme is

$$v_m^{n+1} = v_m^n - \frac{\nu}{2}(3v_m^n - 4v_{m-1}^n + v_{m-2}^n) + \frac{\nu^2}{2}(v_m^n - 2v_{m-1}^n + v_{m-2}^n). \tag{40}$$

Let $q = e^{-\mathrm{i}\theta}$. Then

$$G = 1 - \frac{\nu}{2}(3 - 4q + q^2) + \frac{\nu^2}{2}(1 - 2q + q^2). \tag{41}$$

Set $\delta = 1 - q$. Since

$$3 - 4q + q^2 = 2\delta + \delta^2, \qquad 1 - 2q + q^2 = \delta^2, \tag{42}$$

we get

$$G = 1 - \nu\delta + \frac{\nu(\nu - 1)}{2}\delta^2. \tag{43}$$

Using

$$\delta + \overline{\delta} = |\delta|^2 = 4\sin^2(\theta/2), \tag{44}$$

one obtains

$$|G|^2 = 1 - 4\nu(2 - \nu)(1 - \nu)^2 \sin^4(\theta/2). \tag{45}$$

Thus

$$|G| \le 1 \quad \text{for all } \theta \quad \Longleftrightarrow \quad 0 \le \nu \le 2. \tag{46}$$

4 Implicit Schemes

4.1 BTFS

BTFS is

$$v_m^{n+1} + \nu(v_{m+1}^{n+1} - v_m^{n+1}) = v_m^n. \tag{47}$$

Thus

$$G\left[1 + \nu(e^{\mathrm{i}\theta} - 1)\right] = 1, \tag{48}$$

so

$$G = \frac{1}{1 - \nu + \nu e^{\mathrm{i}\theta}}. \tag{49}$$

The denominator satisfies

$$|1 - \nu + \nu e^{\mathrm{i}\theta}|^2 = 1 - 2\nu(1 - \nu)(1 - \cos\theta). \tag{50}$$

For $0 < \nu < 1$, this can be smaller than 1, so $|G| > 1$ for some modes. For $\nu \geq 1$, the denominator modulus is at least 1. BTFS is therefore not the natural stable implicit upwind scheme for $a > 0$ when $0 < \nu < 1$.

4.2 BTBS

BTBS is

$$v_m^{n+1} + \nu(v_m^{n+1} - v_{m-1}^{n+1}) = v_m^n. \tag{51}$$

Then

$$G\left[1 + \nu - \nu e^{-\mathrm{i}\theta}\right] = 1, \tag{52}$$

and

$$G = \frac{1}{1 + \nu - \nu e^{-\mathrm{i}\theta}}. \tag{53}$$

The denominator has modulus squared

$$|1 + \nu - \nu e^{-\mathrm{i}\theta}|^2 = 1 + 2\nu(1 + \nu)(1 - \cos\theta) \geq 1. \tag{54}$$

Hence

$$|G| \leq 1 \qquad \text{for all } \nu \geq 0. \tag{55}$$

BTBS is unconditionally stable for right-moving advection.

4.3 BTCS

BTCS is

$$v_m^{n+1} + \frac{\nu}{2}(v_{m+1}^{n+1} - v_{m-1}^{n+1}) = v_m^n. \tag{56}$$

Therefore

$$G\left[1 + \frac{\nu}{2}(e^{\mathrm{i}\theta} - e^{-\mathrm{i}\theta})\right] = 1, \tag{57}$$

or

$$G = \frac{1}{1 + \mathrm{i}\nu\sin\theta}. \tag{58}$$

Thus

$$|G|^2 = \frac{1}{1+\nu^2\sin^2\theta} \le 1. \tag{59}$$

BTCS is unconditionally stable in the von Neumann sense.

4.4 Crank–Nicolson

Crank–Nicolson is

$$v_m^{n+1} + \frac{\nu}{4}(v_{m+1}^{n+1} - v_{m-1}^{n+1}) = v_m^n - \frac{\nu}{4}(v_{m+1}^n - v_{m-1}^n). \tag{60}$$

Substitution gives

$$G\left[1 + \frac{\mathrm{i}\nu}{2}\sin\theta\right] = 1 - \frac{\mathrm{i}\nu}{2}\sin\theta. \tag{61}$$

Hence

$$G = \frac{1 - \frac{\mathrm{i}\nu}{2}\sin\theta}{1 + \frac{\mathrm{i}\nu}{2}\sin\theta}. \tag{62}$$

The numerator and denominator are complex conjugates, so

$$|G| = 1. \tag{63}$$

Crank–Nicolson is neutrally stable for all ν.

5 Interpreting the λ Parameter Sweeps

The numerical parameter sweep in the Beamer slides is a direct computational test of the stability conditions derived above. In the assignment problem the equation is

$$u_t + u_x = 0, \tag{64}$$

so the wave speed is $a = 1$. Therefore

$$\nu = \frac{ak}{h} = \frac{k}{h} = \lambda. \tag{65}$$

Thus the parameter λ is exactly the Courant number. It measures the distance travelled by the exact characteristic during one time step, measured in grid cells:

$$\lambda = \frac{\text{physical distance travelled in one time step}}{\text{grid spacing}} = \frac{ak}{h}. \tag{66}$$

Since $a = 1$, after one time step the exact solution shifts by a distance k, or by $k/h = \lambda$ grid cells.

5.1 Why $\lambda = 1$ is the Important Threshold

For the exact advection equation,

$$u(x_m, t_{n+1}) = u(x_m - k, t_n). \tag{67}$$

On the grid, this means the foot of the characteristic lies at

$$x_m - k = x_m - \lambda h. \tag{68}$$

If $0 \le \lambda \le 1$, the foot of the characteristic lies between x_m and the immediate upwind grid point x_{m-1}. This is why one-cell explicit upwind-type methods can be stable in this range.

If $\lambda > 1$, the exact characteristic foot lies farther than one grid cell upstream:

$$x_m - k < x_{m-1}. \tag{69}$$

A one-step explicit method whose numerical domain of dependence only reaches one grid cell cannot correctly contain the physical domain of dependence. This is the Courant–Friedrichs–Lewy, or CFL, obstruction. It is not merely an accuracy issue: for many explicit schemes it appears as von Neumann instability.

This agrees with the amplification-factor calculations:

$$\text{FTBS, CTCS, Lax–Friedrichs, Lax–Wendroff} \quad \Longrightarrow \quad |\lambda| \le 1. \tag{70}$$

Beam–Warming is different because it uses a two-point upwind-biased stencil involving v_m^n, v_{m-1}^n, and v_{m-2}^n. Its numerical domain of dependence reaches two upwind cells, and the von Neumann calculation gives the wider stability range

$$0 \le \lambda \le 2. \tag{71}$$

The implicit schemes solve a global linear system at each time step, so their numerical domain of dependence is not limited to a fixed explicit stencil. This is why BTBS, BTCS, and Crank–Nicolson remain stable in the von Neumann sense for all tested λ values. Stability, however, does not automatically mean high accuracy.

5.2 Quantities Plotted in the Sweep

The sweep plots use

$$b = 3, \qquad h = 0.01, \qquad T = 1.5, \qquad \lambda \in \{0.8, 0.9, 1.0, 1.01, 1.05, 1.1\}. \tag{72}$$

For each method, two quantities are plotted on a logarithmic vertical scale:

$$\left\|e^N\right\|_\infty = \max_m |v_m^N - u(x_m, T)|, \qquad \left\|v^N\right\|_\infty = \max_m |v_m^N|. \tag{73}$$

The first quantity measures the maximum pointwise error at the final time. The second quantity measures the maximum numerical amplitude. The amplitude is a useful stability diagnostic because the exact solution satisfies

$$0 \le u(x,t) \le 1, \qquad \|u(\cdot,t)\|_\infty = 1 \tag{74}$$

as long as the pulse is inside the domain. Therefore, if a computed solution has $\left\|v^N\right\|_\infty$ of size 10^3, 10^8, or larger, that growth cannot be physical. It is evidence of numerical instability.

The right boundary is not responsible for the observed instability in this sweep. The initial pulse has support in $[-1, 1]$, and at $T = 1.5$ the exact pulse has support approximately

$$[-1 + 1.5,\ 1 + 1.5] = [0.5, 2.5]. \tag{75}$$

Thus the outflow boundary $b = 3$ is still to the right of the pulse. Changing b from 3 to 4 or 5 should not significantly change the error, and this is exactly what the boundary sweep confirms.

5.3 CFL-Limited Explicit Methods

The first split plot groups the schemes whose theoretical stability range is governed by the threshold $\lambda = 1$.

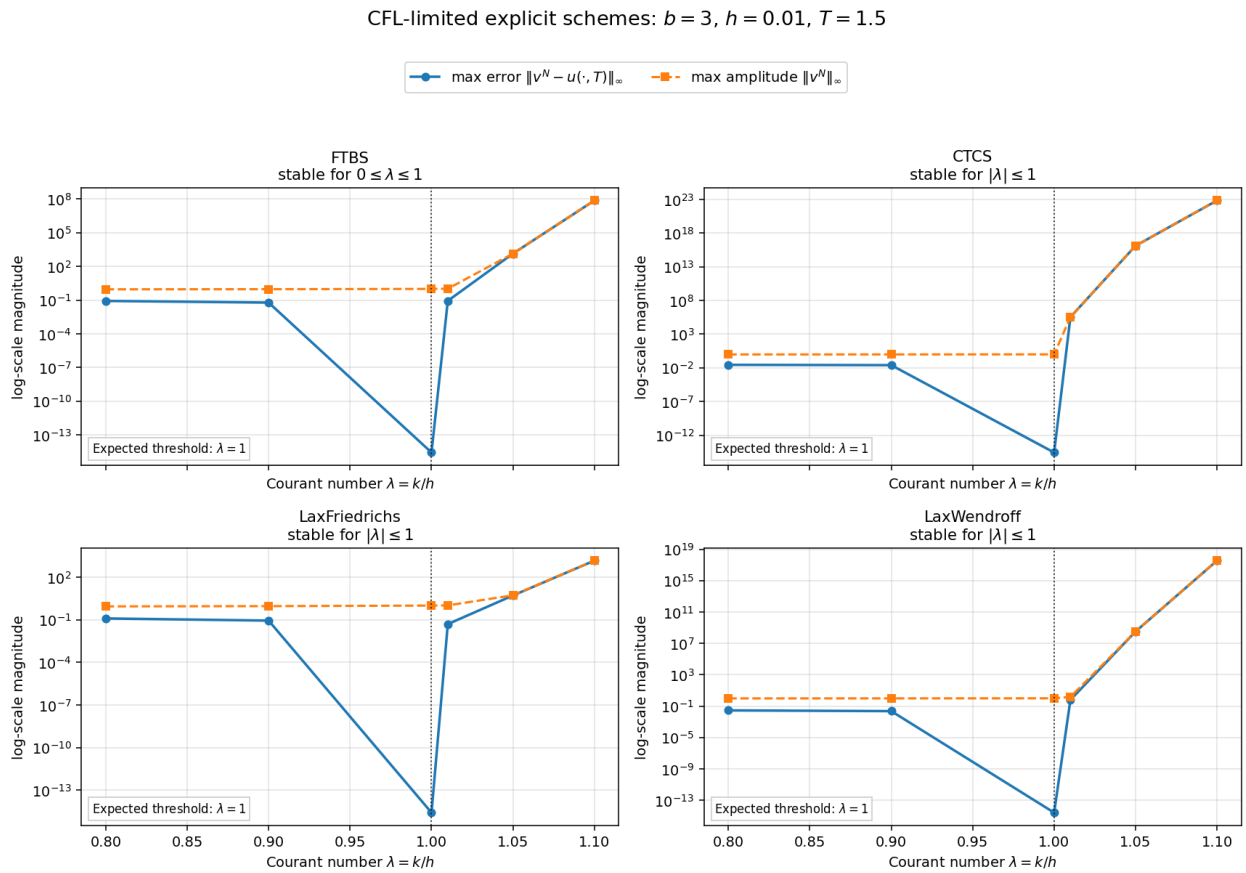

Figure 1: Sweep in λ for explicit schemes with CFL threshold $\lambda = 1$.

For FTBS, CTCS, Lax–Friedrichs, and Lax–Wendroff, the derived stability condition is

$$|\lambda| \le 1. \tag{76}$$

The plot is consistent with this theory:

- For $\lambda = 0.8$ and $\lambda = 0.9$, the maximum amplitude remains of order one.
- At $\lambda = 1$, several schemes become nearly exact in this special experiment because the wave moves exactly one grid cell per time step.
- Just above the threshold, such as $\lambda = 1.01$, instability may not be dramatic over a finite time interval because the unstable amplification factor is only slightly larger than one.
- For $\lambda = 1.05$ and $\lambda = 1.1$, the unstable modes have had enough growth to become plainly visible, and the maximum amplitude increases rapidly.

The near-zero error at $\lambda = 1$ should not be interpreted as a general convergence result. It is a special grid-alignment effect. For example, FTBS becomes

$$v_m^{n+1} = v_{m-1}^n \qquad \text{when } \lambda = 1, \tag{77}$$

which is exactly the grid shift produced by the continuous equation over one time step. Similar exact-shift behavior occurs for Lax–Friedrichs and Lax–Wendroff at $\lambda = 1$.

5.4 Unstable, Wider-Range, and Implicit Methods

The second split plot groups the methods that do not follow the simple $|\lambda| \leq 1$ story.

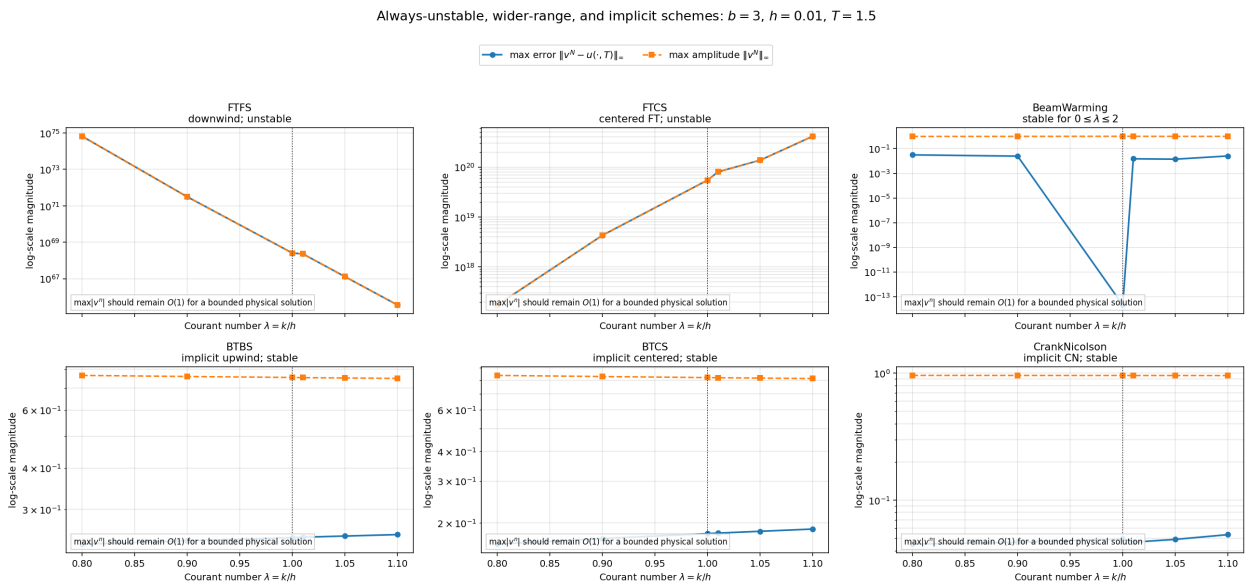

Figure 2: Sweep in λ for always-unstable, wider-range, and implicit schemes.

FTFS is unstable because it is the downwind method for the right-moving equation $u_t + u_x = 0$. Its amplification factor satisfies

$$|G|^2 = 1 + 2\lambda(1 + \lambda)(1 - \cos\theta) > 1 \tag{78}$$

for nonzero modes. Thus reducing λ below 1 does not fix FTFS; the spatial bias is wrong.

FTCS is also unstable for every nonzero λ, since

$$|G|^2 = 1 + \lambda^2 \sin^2\theta > 1 \tag{79}$$

for many Fourier modes. Hence the FTCS curve shows large growth even when $\lambda = 0.8$.

Beam–Warming is stable over the tested range because the theory gives

$$0 \leq \lambda \leq 2. \tag{80}$$

The tested values only go up to $\lambda = 1.1$, so the method remains bounded in the sweep. This does not mean Beam–Warming is always monotone or nonoscillatory; it is a higher-order upwind method and can still produce dispersive errors near nonsmooth features.

The implicit methods BTBS, BTCS, and Crank–Nicolson remain bounded as λ increases. Their different behavior is visible in the amplitude:

- BTBS is implicit upwind and strongly dissipative, so $\left\|v^N\right\|_\infty$ is noticeably below 1.
- BTCS is also stable but still introduces smoothing and phase error.
- Crank–Nicolson is neutrally stable, $|G| = 1$, so it preserves amplitudes better, but it may retain dispersive oscillations near nonsmooth corners.

5.5 Main Conclusion from the Sweep

The λ sweep should be read as numerical evidence for the theoretical amplification-factor conditions:

$$\boxed{\text{Stability is controlled by } |G(\theta)| \leq 1.} \tag{81}$$

For CFL-limited explicit schemes, the threshold $\lambda = 1$ is visible. For FTFS and FTCS, instability occurs even below $\lambda = 1$ because their amplification factors already exceed one. For Beam–Warming, the wider range $0 \leq \lambda \leq 2$ explains why the method stays bounded beyond $\lambda = 1$. For implicit methods, the absence of a CFL stability restriction explains why the amplitude remains bounded, although accuracy and numerical diffusion still depend on h and k.

6 Summary

Scheme	Amplification factor	Stability condition
FTFS	$1 + \nu - \nu e^{\mathrm{i}\theta}$	unstable for $\nu > 0$
FTBS	$1 - \nu + \nu e^{-\mathrm{i}\theta}$	$0 \leq \nu \leq 1$
FTCS	$1 - \mathrm{i}\nu \sin\theta$	unstable for $\nu \neq 0$
CTCS	$G^2 + 2\mathrm{i}\nu \sin\theta\, G - 1 = 0$	$\lvert\nu\rvert \leq 1$
Lax–Friedrichs	$\cos\theta - \mathrm{i}\nu \sin\theta$	$\lvert\nu\rvert \leq 1$
Lax–Wendroff	$1 - \mathrm{i}\nu \sin\theta + \nu^2(\cos\theta - 1)$	$\lvert\nu\rvert \leq 1$
Beam–Warming	second-order upwind symbol	$0 \leq \nu \leq 2$
BTFS	$(1 - \nu + \nu e^{\mathrm{i}\theta})^{-1}$	stable for $\nu \geq 1$ on a periodic grid
BTBS	$(1 + \nu - \nu e^{-\mathrm{i}\theta})^{-1}$	all $\nu \geq 0$
BTCS	$(1 + \mathrm{i}\nu \sin\theta)^{-1}$	all ν
Crank–Nicolson	$\dfrac{1 - \frac{\mathrm{i}\nu}{2}\sin\theta}{1 + \frac{\mathrm{i}\nu}{2}\sin\theta}$	all ν, $\lvert G\rvert = 1$